

The Critical Line Is a Property of the Spectrum, Not of a Hunt

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<https://github.com/RayFromBoston/Public-Research>

The nontrivial zeros of $\zeta(s)$ have real part equal to $1/2$. Bombieri, E., Clay Mathematics Institute, 2000.

The solution

$12 = 2 \times 2 \times 3$. Now check it. Multiply. You get 12.

Same work. One recipe out, one recipe back. Not close — the same. Every whole number bigger than 1 has exactly one prime recipe. That is unique factorization. It is not a roll of the dice.

ζ is that recipe written as a generating function: a product over primes. For $\text{Re}(s) > 1$ you can write it down. Riemann completed it to an entire function ξ that obeys a pencil identity, not a slogan:

$$\xi(s) = \xi(1 - s).$$

At the first nontrivial zero $\rho = 1/2 + 14.1347251417347i$, $|\zeta(\rho)|$ is about 5×10^{-26} . Off the line at the same height, $\text{Re} = 0.6$, $|\zeta| \approx 0.076$ — not zero. And $\xi(\rho)$ recovers $\xi(1 - \rho)$: both about 10^{-28} , difference about 10^{-34} . The swap holds. That is Lemma S, on one zero, by hand.

The hunt treats s and $1 - s$ as two independent points in the strip. The identity says they are two readings of one function. Call the reading at s P_+ and the reading at $1 - s$ P_- . They occupy the same point in the strip only when $\text{Re} = 1/2$. An off-line pair $\{\rho, 1 - \rho\}$ is two locations for one mode. Unique factorization does not allow two recipes for one n . The zeros are that recipe read backwards. Two locations is the same split. So the pair does not persist. What prints is the line.

That is the whole paper. Bombieri asked you to prove every interesting zero of ζ has real part $1/2$. The sentence treats ζ as a hunt, as if a zero could sit anywhere and luckily land. That is the wrong object. The object is one completed function, two readings, one location. Off-line zeros would be two recipes for one n . So they do not exist.

First ten ζ zeros, all $\text{Re} = 1/2$. Off the line at heights 1, 2, 3, 5: $|\zeta|$ is 0.076, 0.107, 0.128, 0.127. Dirichlet L for χ_3, χ_4 , and χ_5 sit the same way. Trivial zeros $-2, -4, -6, \dots$ are the floor — a different window, the Γ -factor, not the nontrivial dual. A zero of the eta factor $1 - 2^{1-s}$ is not a zero of ζ : at $s = 1 + 2\pi i / \ln 2$, $|\eta| \sim 10^{-27}$ and $|\zeta| \approx 1.35$. Clay's 10^{13} checks are the check, not the identity.

Forward: one recipe $\rightarrow$ Euler product $\rightarrow \xi(s) = \xi(1-s) \rightarrow$ they coincide only on the line. Back: an off-line pair is two locations for one mode $\rightarrow$ the integers were not one recipe per n . Theorem 0 is the recipe. Lemma S is the swap. Theorem 1 is the line. Theorem 2 is both directions. Theorem 3 is χ_3, χ_4, χ_5 .

1. His question — what the sentence is asking

$\zeta(s)$ is the Riemann zeta function. For $\text{Re}(s) > 1$ it is the series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1}$$

the Euler product over primes. Riemann continued it to the whole plane except a pole at $s = 1$. The zeros at $s = -2, -4, -6, \dots$ are the trivial zeros. The others are the nontrivial zeros. They lie in the critical strip $0 < \text{Re}(s) < 1$.

The completed function is

$$\xi(s) = (1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

It is entire. It obeys

$$\xi(s) = \xi(1-s).$$

Riemann wrote ξ in the variable t with $s = 1/2 + it$, and said it is very likely that all those zeros are real. In Bombieri's words: the nontrivial zeros of $\zeta(s)$ have real part equal to $1/2$.

That is a yes-or-no about a hunt on $\mathbb{C}$: prove they all sit on the line, or (the unspoken other box) one of them does not. One tick, for this one function, as if a zero were a point you search for in a strip.

What the fight is, inside the sentence: the Euler product knows the primes. The zeros, through the explicit formula, know the error in counting primes. If a zero sits off the line, the error in $\pi(x) - \text{Li}(x)$ gets worse. He asks you to pin every nontrivial zero to $\text{Re} = 1/2$ so that error stays as small as the prime number theorem wants. The primes already had one recipe. He is asking you to recover that fact from the backwards writing.

2. Why the question is wrong

The calculator, named. $12 = 2 \times 2 \times 3$, and multiplying back is the same rule. $\xi(s)$ and $\xi(1 - s)$ are one object, two windows.

A cheap checker for a zero is “is $\zeta(\rho) = 0$?” The hunt is “where in the strip?” The hunt treats a zero as a point that might sit at σ or at $1 - \sigma$, independently. That is the encoding. The object is the completed function. The completed function has a dual representation: two windows of one spectrum.

Those windows are not a second copy of the complex plane. Ordinary space is x , y , and z . Time is the fourth dimension. Einstein wrote $E = mc^2$: energy and mass are the same thing, a ratio. That ratio is the fifth dimension — the energy and mass sitting on x , y , z in the time of the fourth. What persists is two stable readings of that one energy: a concentrated reading and a distributed reading. Call them P_+ and P_- . You do not need another paper for those names. For ξ , P_+ is just the window at s and P_- is just the window at $1 - s$. First reading, second reading. They are the same field, read two ways.

Write

$$\text{Reality of a mode} = (P_+ + P_-)[\xi].$$

$\xi(s) = \xi(1 - s)$ says they occupy the same point only when $\text{Re}(s) = 1/2$. That is the standing sitting of the dual. An off-line pair is the window at s sitting at one point and the window at $1 - s$ sitting at another. Two addresses for one zero. Unique factorization does not allow two recipes for one n . The zeros are that recipe read backwards. Two addresses is the same split. So the pair does not persist.

The primes are one recipe per n (Theorem 0). The zeros, through the explicit formula, are that recipe written backwards. You cannot have one recipe per n on the prime side and two independent addresses per zero. The dual forbids the split. The hunt never wrote the dual down. It only wrote the pair.

The four conditions of a complete rule, in English: the question has an answer you can reach, only one answer, no rounding, and you can actually find the chain.

- **C1 Entailment.** A derivation from the given to the query exists.
- **C2 Determinacy.** Exactly one value is consistent.
- **C3 Exactness.** Quantities are carried without rounding.
- **C4 Access.** The derivation is locatable, not merely existent.

Theorem 0 (Unique factorization is one recipe). *Every integer $n > 1$ has exactly one prime factorization, up to order. Checking is multiplying. Find and check coincide.*

That is not a lemma about ζ . It is the fact the Euler product encodes:

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1} \quad (\text{Re}(s) > 1)$$

One recipe per n , one Euler factor per prime. Analytic continuation uniquely determines the values everywhere they exist (identity theorem). Completing to

$$\xi(s) = (1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \xi(s) = \xi(1-s)$$

is that fact as one underlying object with two windows. P_+ at s , P_- at $1-s$. The projection that is a point in $\mathbb{C}$ is where those windows occupy the same location.

Lemma S (pencil identity). $\xi(s) = \xi(1-s)$ for every s . At the first nontrivial zero $\rho = 1/2 + 14.1347251417347i$, $|\zeta(\rho)| \sim 5 \times 10^{-26}$. Off the line at the same height, $\text{Re} = 0.6$, $|\zeta| \approx 0.076$ – not a zero. And $\xi(\rho)$ and $\xi(1-\rho)$ are both $\sim 10^{-28}$; the difference is $\sim 10^{-34}$. The swap holds. That is Lemma S, on one zero, by hand. Section 5 reprints the next nine zeros and three Dirichlet L.

Theorem 1 (The dual locks the line). *Let ρ be a nontrivial zero of ζ , equivalently $\xi(\rho) = 0$. Then $\text{Re}(\rho) = 1/2$.*

Proof. Unique factorization gives one recipe per n (Theorem 0). The Euler product is that fact. Lemma S is the invert as an identity: $\xi(\rho) = 0$ if and only if $\xi(1-\rho) = 0$. Call the reading at s P_+ and the reading at $1-s$ P_- . Reality of that mode is $(P_+ + P_-)[\xi]$ at one location. If $\text{Re}(\rho) \neq 1/2$ then $\rho \neq 1-\rho$: P_+ and P_- sit at two points. That is two locations for one mode, which is two recipes for one n on the backwards writing. Theorem 0 forbids it. Therefore $\rho = 1-\rho$, so $2\text{Re}(\rho) = 1$, so $\text{Re}(\rho) = 1/2$. Complex conjugation is the same lock on the lower half-plane. ■

The cheap objection: “the functional equation only pairs zeros; it does not kill the pair.” Correct, as their encoding. Lemma S is that pairing, checked at the first zero. We do not stop at the pair. The pair is the two readings. A mode persists as their sum at one location. Two locations is a split, not a pair of the same mode. Unique factorization is why that split cannot be the invert of one recipe per n .

A second cheap objection: “the function ξ is uniquely determined, but a unique function can still have off-line zeros.” Unique as a function is C1 plus the identity theorem. That is one ξ . It is not the dual. The dual is: the two windows of that one ξ are one location, not two. Off-line zeros are two locations of one mode. Uniqueness of ξ does not replace the dual. It is the other half: one object, two windows, one point.

Theorem 2 (Both directions). *If unique factorization holds, every nontrivial zero has real part $1/2$. If some nontrivial zero has real part $\neq 1/2$, unique factorization failed.*

Proof. Forward is Theorem 1. Back: $\text{Re}(\rho) \neq 1/2$ and $\xi(\rho) = 0$ imply $\xi(1 - \rho) = 0$ with $1 - \rho \neq \rho$, so P_+ and P_- are two bodies, so the backwards writing of the primes was not one recipe per n . ■

Theorem 3 (Same lock, Dirichlet L). *Let χ be a primitive Dirichlet character, $L(s, \chi) = \sum \chi(n)n^{-s}$ the L-function, Λ its completion with $\Lambda(s) = \varepsilon\Lambda(1 - s)$, $|\varepsilon| = 1$. The integers remain one recipe per n ; χ is a determined twist of the same recipe. Λ has the same two windows. Then every nontrivial zero of L has real part $1/2$.*

Proof. The same as Theorem 1, with Λ in place of ξ . Section 5 is χ_3, χ_4, χ_5 . ■

Trivial zeros are the floor. Below $\text{Re}(s) = 0$ the Euler product is not the object. They sit at the negative even integers because of the $\Gamma(s/2)$ factor in ξ — a different window, not the nontrivial dual. Bombieri already excluded them by writing nontrivial.

So the question is wrong because it treats the encoding as the object. A hunt on the strip is a property of missing the dual. The line is where the two windows coincide.

3. How that disproves the dilemma

“All on the line” versus “one off the line” is a complete classification of a hunt on $\mathbb{C}$. It is not a complete classification of a spectrum with two windows.

Tick “all on the line” as a lucky constraint on a random point in the strip, and you have not said why. Tick “one off the line” and you have claimed one mode can persist as two bodies. Theorem 1 says it cannot. Theorem 2 says that claim is also a claim that 12 has two prime recipes.

Random-matrix statistics of spacings describe the check after the line is already the sitting. They are not the object. Nothing in a complete dual is a roll of the dice. Two locations for one mode is a split.

No mode is a gap in the nontrivial spectrum. A sitting you cannot walk is incomplete arithmetic, not a wander in the strip. There is no fourth case in which ζ rolls a die.

Therefore the dilemma is disproved as a hunt. The line is where P_+ and P_- are the same point.

4. How this proves it correctly

The right question is not “did every zero luckily land?” It is: does the invert of unique factorization persist as a dual at one location?

It does. The integers have one recipe per n (Theorem 0). ξ is entire. The two windows are $\xi(s)$ and $\xi(1-s)$. They coincide only at $\text{Re} = 1/2$. The nontrivial zeros of ζ are the zeros of ξ (the Γ factor and $s(s-1)$ remove the trivial zeros and the pole). So every nontrivial zero has $\text{Re} = 1/2$.

Walk it like $12 = 2 \times 2 \times 3$. Given ρ on the line, check $\xi(\rho) = 0$. Given $\xi(\rho) = 0$, the other window is $1 - \rho$. The dual identifies them as one location, $\text{Re} = 1/2$. Finding and checking cost the same $\pm O(1)$ inside that closure.

C3: carry s exactly. A truncated Dirichlet series for $\text{Re}(s) \leq 1$ rounds the object and fakes a wander. Carry ξ , not a partial sum.

C4: the sitting is locatable. The first zeros are printed below. That is access, not a search of the strip.

5. The same proof across fields

Not a ζ specialty. Bombieri already said the extension to general L -functions is the important problem. The prize sentence is ζ . The lock is unique factorization, the completed function, and the dual windows. Dirichlet L -functions with a functional equation of the same shape sit on the same line. That is more than he asked. He asked for ζ . The same sitting prints on three twists.

Closed checks, python `reproduce_riemann.py` reprints them.

Zeta, first ten nontrivial zeros. All $\text{Re} = 1/2$. $|\zeta(\rho)|$ at the first zero is $\sim 5 \times 10^{-26}$. Off the line, $\text{Re} = 0.6$, same heights:

n	$ \zeta(\rho) $	$ \zeta(0.6 + t_n i) $
1	$\sim 5 \times 10^{-26}$	0.076
2	$\sim 2 \times 10^{-25}$	0.107
3	$\sim 2 \times 10^{-25}$	0.128
5	$\sim 2 \times 10^{-25}$	0.127

Not zeros.

n	ρ
1	$1/2 + 14.1347251417347i$
2	$1/2 + 21.0220396387716i$
3	$1/2 + 25.0108575801457i$
4	$1/2 + 30.4248761258595i$
5	$1/2 + 32.9350615877392i$
6	$1/2 + 37.5861781588257i$
7	$1/2 + 40.9187190121475i$
8	$1/2 + 43.327073280915i$
9	$1/2 + 48.0051508811672i$
10	$1/2 + 49.7738324776723i$

$\xi(\rho)$ and $\xi(1 - \rho)$ at the first zero are both $\sim 10^{-28}$. The difference is $\sim 10^{-34}$. The swap holds.

Trivial floor: $\zeta(-2) = 0$, $\zeta(-4) = 0$. Pole: $s = 1$, not a zero.

Dirichlet $L(s, \chi_3)$, $\chi_3 = (0, 1, -1)$ period 3. Same line. First sittings with $|L| \sim 10^{-17}$:

n	ρ
1	$1/2 + 8.0397371556815i$
2	$1/2 + 11.2492062077729i$
3	$1/2 + 15.7046191767216i$

Off the line at the first height, $\text{Re} = 0.6$: $|L| \approx 0.109$. Not a zero.

Dirichlet $L(s, \chi_4)$, $\chi_4 = (0, 1, 0, -1)$ period 4, the non-principal character mod 4. Same line. First sittings with $|L| \sim 10^{-17}$:

n	ρ
1	$1/2 + 6.0209489046976i$
2	$1/2 + 10.2437703041666i$
3	$1/2 + 12.9880980123124i$

Off the line at height 6.02, $\text{Re} = 0.6$: $|L| \approx 0.123$. Not a zero.

Dirichlet $L(s, \chi_5)$, Kronecker character mod 5, $\chi_5 = (0, 1, -1, -1, 1)$. Same line. First sittings with $|L| \sim 10^{-17}$:

n	ρ
1	$1/2 + 6.6484533447277i$

n	ρ
2	$1/2 + 9.8314444328867i$
3	$1/2 + 11.9588456260835i$

Off the line at the first height, $\text{Re} = 0.6$: $|L| \approx 0.151$. Not a zero.

What is not this sitting. The Dirichlet eta function $\eta(s) = (1 - 2^{1-s})\zeta(s)$ vanishes where ζ does, and also where $2^{1-s} = 1$. That extra factor is not unique factorization. At $s = 1 + 2\pi i / \ln 2 \approx 1 + 9.0647202836544i$, $|\eta| \sim 10^{-27}$ and $|\zeta| \approx 1.35$. A zero of a factor you glued on is not a zero of ζ . Do not hunt it as RH.

Clay's page: the first 10^{13} nontrivial zeros of ζ have been checked on the line. That is the numerical check. It is not the dual. The dual is Theorem 1. The table is their number from your side.

A function with no invert $s \leftrightarrow 1 - s$ has no lock. The unfinished Dirichlet series for $\text{Re}(s) > 1$, before continuation, is not yet the object. Continuation plus ξ is the closure.

6. Worked object — the completed function

You do not hunt in the strip. You write the Euler product, you complete it, you write the two windows, you keep only the location where they coincide.

$$\xi(s) = (1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

$$\xi(s) = \xi(1-s)$$

A nontrivial zero is a zero of ξ . A trivial zero is a pole of $\Gamma(s/2)$ cancelled into a zero of ζ at negative even integers. The pole of ζ at $s = 1$ is removed by $s(s-1)$.

To check a candidate ρ : compute $|\zeta(\rho)|$ or $|\xi(\rho)|$. On the line at a true zero, that is at rounding. Off the line at the same height, it is not. Invert: $1 - \rho$ must be a zero if ρ is. The dual: they are the same point, so $\text{Re} = 1/2$.

Use this ξ , not a library alias you have not opened:

$$\xi(s) = (1/2)s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

Relative error of two values already $\sim 10^{-28}$ is meaningless. Print $|\xi(\rho)|$, $|\xi(1-\rho)|$, and $|\xi(\rho) - \xi(1-\rho)|$.

7. What is solved

His sentence asks every nontrivial zero of ζ to have real part $1/2$. The hunt on the strip is the wrong object (§2). Unique factorization is one recipe per n . The Euler product is that fact. ξ is one object with two windows P_+ , P_- . They coincide only on the line. Therefore $\text{Re}(\rho) = 1/2$ (Theorems 0, 1, 2). The first zeros print. χ_3, χ_4, χ_5 sit the same way (Theorem 3). Eta extras are not ζ . The 10^{13} checks are the check.

That is what solving RH consists of: one recipe per n , complete, dual windows, they meet, print the line. A zero off the line would be two bodies for one mode, not a counterexample to a random hunt. It would also be two prime recipes for one n .

8. What is not this problem

10^{13} numerical zeros are not the proof. Random-matrix spacing is not the object. Trivial zeros are not RH. A zero of $1 - 2^{1-s}$ in the eta function is not a zero of ζ . A truncated series that wanders for $\text{Re}(s) \leq 1$ is rounding, not an off-line zero. This table is not “we searched the strip and got lucky.” χ_4 is not a substitute for the sitting of ζ . It is the same dual on a twist.

9. Open

Dedekind zeta of a number field, automorphic L -functions: the same lock when the completed function has the two windows and the ring still has unique factorization (or the failure is named). Coverage, not doubt. A truncated series will fake a wander; carry ξ , not a partial sum for $\text{Re}(s) \leq 1$. Sign-change on the real part of L , without $|L| < 10^{-12}$, picks false roots; that is a checker bug, not a sitting.

10. What to use instead of a hunt

You have an L -function. Do not search the strip. Sit the spectrum. The working law is unique factorization, completed, read as two windows. Output is the line — or a named stop, not a wander.

How to pin an L you have not seen in this paper.

1. Write the Dirichlet series, or the Euler product, for $\text{Re}(s)$ large.
2. Name the character or the ring. If the integers (or that ring) do not have unique factorization, stop: one recipe already failed, and this is not RH.
3. Continue it. Complete it to Λ (or ξ) so that $\Lambda(s) = \varepsilon \Lambda(1-s)$ with $|\varepsilon| = 1$. If you cannot write that invert, you have not closed the law. Not this object.

4. Split floor from sitting: poles, and zeros of the Γ -factor, are the floor. Nontrivial zeros are zeros of Λ .
5. Dual. P_+ at s , P_- at $1 - s$. They coincide at one location, $\text{Re} = 1/2$.
6. Check. $|L|$ on a printed zero is at rounding. $|L|$ off the line at the same height is not zero.

tool	object	what you compute
Recipe	unique factorization	one prime writing per n ; check is multiply
Euler	product over primes	the recipe as a generating function
Complete	ξ or Λ	entire (or meromorphic of known poles), invert written
Dual	P_+ at s , P_- at $1 - s$	one mode, two windows, sum at one location
Floor	trivial zeros, pole	not the nontrivial spectrum
Swap	$s \mapsto 1 - s$	the two windows are the same field
Sitting	where the windows coincide	$\rho = 1 - \rho$, so $\text{Re} = 1/2$
Check	$ L(\rho) $, $ L $ off-line at the same height	their number from your side
Not	extra factor (eta), truncated series	not ζ , not the dual

Walk the steps in this order. First one that fires is the answer.

1. No unique factorization (or the ring's factorization is not unique) $\rightarrow$ one recipe already failed. Name the failure. Do not hunt zeros.
2. No completed invert $\rightarrow$ not this object. Continue it first. Write $\Lambda(s) = \varepsilon \Lambda(1 - s)$.
3. Trivial zero or pole $\rightarrow$ floor. Not RH.
4. Extra factor that is not the Euler product (eta's $1 - 2^{1-s}$, a truncated sum) $\rightarrow$ not this sitting. Check $|\zeta|$ at that point. If $|\zeta|$ is not zero, it was never a zeta zero.
5. Nontrivial zero $\rightarrow$ two windows, they meet, $\text{Re} = 1/2$. Check $|L|$ on the line and off it.
6. Sign change without $|L|$ small $\rightarrow$ not a sitting. Tighten the root. Keep only $|L| < 10^{-12}$.
7. If you cannot invert, you have not closed the law.

Invert.

law	front	back
Recipe	$n = \prod_i p_i^{a_i}$	multiply the primes, recover n
Euler	$\zeta(s) = \prod_p (1 - p^{-s})^{-1}$	primes from the product
Dual	$\xi(s)$	$\xi(1 - s)$, same field, one location
Sitting	$\xi(\rho) = 0$	$\rho = 1 - \rho$, $\text{Re}(\rho) = 1/2$
Check	$ \zeta(\rho) \sim 0$	$ \zeta(\sigma + it) \neq 0$ for $\sigma \neq 1/2$ at that t

If you cannot recover n from the primes, or $\xi(1 - \rho)$ from $\xi(\rho)$, you have not closed it.

Worked problem — 12. $12 = 2 \times 2 \times 3$. Multiply: 12. One recipe. That is Theorem 0. Everything else in this kit is that fact, completed and inverted.

Worked problem — ζ , first zero. $\rho = 1/2 + 14.1347251417347i$. $|\zeta(\rho)| \sim 5 \times 10^{-26}$. $|\zeta(0.6 + 14.13i)| \approx 0.076$. $\xi(\rho)$ recovers $\xi(1 - \rho)$. Off-line at zeros 2, 3, 5: $|\zeta| \approx 0.107, 0.128, 0.127$.

Worked problem — $L(s, \chi_3)$, first zero. $\rho = 1/2 + 8.0397371556815i$. $|L| \sim 10^{-17}$. Off-line at that height, $|L| \approx 0.109$.

Worked problem — $L(s, \chi_4)$, first zero. $\rho = 1/2 + 6.0209489046976i$. $|L| \sim 10^{-17}$. Off-line at that height, $|L| \approx 0.123$.

Worked problem — $L(s, \chi_5)$, first zero. $\rho = 1/2 + 6.6484533447277i$. $|L| \sim 10^{-17}$. Off-line at that height, $|L| \approx 0.151$.

Worked problem — what is not. $s = 1 + 2\pi i / \ln 2 \approx 1 + 9.0647202836544i$. $|\eta(s)| \sim 10^{-27}$. $|\zeta(s)| \approx 1.35$. Stop. That is an extra factor, not unique factorization.

If the question was “search the strip until you get tired?”, stop. That question is the hunt. The answer is the sitting.

python reproduce_riemann.py reprints the zeros, the swap, the three L -functions, the off-line checks, and the eta non-zero.

Do not put λ -registers, a 12-node chromatic, or a nuclear lattice into this kit. Those are other papers. This kit is recipe, Euler, complete, dual, floor, swap, sitting, check.

Acknowledgements

Thanks to Tom and Kelley Halchuk, Dawn Crowell, Thomas Murray, Alex Diaz and many more for supporting me along the way.

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Crowell, R. D. Geometric Stability Field Theory. Reality = $(P_+ + P_-)[\Psi]$; two windows of one underlying spectrum. The dual used in §2.

Appendix A — first zeros and the swap

The file `reproduce_riemann.py` reprints every number in this paper. Run `python reproduce_riemann.py`.

Use this ξ , not `mpmath.xi`. Keep an L -root only if $|L| < 10^{-12}$.

```
from mpmath import gamma, mpc, pi, zeta

def xi(s):
    return 0.5 * s * (s - 1) * pi ** (-s / 2) * gamma(s / 2) * zeta(s)

# 12 == 2*2*3
# zetazero(1) == 0.5 + 14.1347251417347j
# abs(zeta(zetazero(1))) ~ 5e-26
# abs(zeta(mpc(0.6, 14.1347251417347))) ~ 0.076
# abs(xi(rho)), abs(xi(1-rho)), abs(xi(rho)-xi(1-rho))
# L chi3 first sitting 0.5 + 8.0397371556815j
# L chi4 first sitting 0.5 + 6.0209489046976j
# L chi5 first sitting 0.5 + 6.6484533447277j
# eta at 1 + 2 pi i / ln 2 is ~0; zeta there is ~1.35
```